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10 September 1986
SUPERSEDING
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9 March 1964
(See 6.7)

MILITARY SPECIFICATION

ESCAPE APPLIANCE, HOODED, AIR INFLATABLE, WITH POUCH, SUBMARINE PERSONNEL

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hooded air-inflated escape appliance with pouch of coated nylon fabric construction and equipped with two pressure relief valves.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-F-106 - Fasteners, Slide; Interlocking.
V-T-295 - Thread, Nylon.
MMM-A-139 - Adhesive, Natural or Synthetic-Natural Rubber.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-L-3454 - Life Preservers; Packaging of.
MIL-W-4088 - Webbing, Textile, Woven Nylon.
MIL-A-5540 - Adhesive, Polychloroprene.
MIL-F-10884 - Fasteners, Snap.
MIL-C-17415 - Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use.
MIL-P-18080 - Plastic Sheets, Vinyl, Flexible, Transparent, Optical Quality.
MIL-C-19002 - Cloth, Coated, and Strip, Coated Cloth - Polychloroprene on Nylon.
MIL-W-27265 - Webbing, Textile, Woven Nylon Impregnated.
MIL-O-81375 - Oral Inflation Assemblies, Survival Equipment, Inflatable.

STANDARDS

FEDERAL

FED-STD-595 - Colors.
FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-1895218 - Escape Appliance, (Hooded) Air Inflatable, With Pouch (for Use by Submarine Personnel); Assembly.
803-1895219 - Escape Appliance, (Hooded) Air Inflatable, With Pouch (for Use by Submarine Personnel); Pouch Assembly and Details.
803-1895220 - Escape Appliance, (Hooded) Air Inflatable, With Pouch (for Use by Submarine Personnel); Belt Assembly and Details.
803-1895221 - Escape Appliance, (Hooded) Air Inflatable, With Pouch (for Use by Submarine Personnel); Development of Details.
803-1895222 - Escape Appliance, (Hooded) Air Inflatable, With Pouch (for Use by Submarine Personnel); Miscellaneous Details.
803-1895223 - Escape Appliance, (Hooded) Air Inflatable, With Pouch (for Use by Submarine Personnel); Hood and Details.
803-1895224 - Pressure Relief Valve Assembly and Details.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 117 - Standard Method of Salt Spray (Fog) Testing.
(DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race, Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Escape appliances furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Material. In instances where the drawings referenced herein do not agree with the materials or design requirements of this specification, the requirements of this specification shall govern.

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3.3.1 Coated cloth. The coated cloth used in the construction of the escape appliance shall conform to type I of MIL-C-19002. The coated cloth tape used in the construction of the escape appliance shall conform to type III of MIL-C-19002. The coated cloth used in the construction of the pouch shall conform to type 2, class A of MIL-C-17415; the requirement for hydrostatic resistance and resistance to accelerated weathering shall not apply. The cloth used in the construction of the neck seal shall be neoprene coated nylon stretch cloth and shall conform to table I.

TABLE I. Coated nylon stretch cloth.

Physical characteristics	
Weight (ounces/sq yd, max)	17.6
Thickness of coating (inch)	0.015 $\pm$ 0.0 - 0.003
Ultimate elongation (percent min)	
(a) Wales direction	80
(b) Courses direction	120
Hydrostatic resistance (pounds per square inch (lb/in^2) min)	50
Appearance	Free of holes and creases; odor not objectionable

3.3.2 Webbing. The dyed nylon webbing used in the escape appliance harness shall conform to table II.

TABLE II. Webbing requirements.

Part	Width	Specification
	(inches)	
Buckle strap	2	MIL-W-4088, type XXIV, impregnated with class L of MIL-W-27265
Back pad	3	MIL-W-4088, type IV
Toggle pouch	3	MIL-W-4088, type IV

3.3.3 Oral inflation valve. The oral inflation valve shall conform to type I of MIL-O-81375, except that the weight shall be 30.0 grams maximum and the valve shall withstand inflation by a blow gun having a discharge nozzle not exceeding 1/8 inch diameter when the gun is connected to a 225 pounds per square inch (lb/in^2) pressure source.

3.3.4 Oral inflation valve clamp. The clamp used to secure the oral valve to the connector tube shall conform to Drawing 803-1895222.

3.3.5 Pressure relief valve. Each escape appliance shall be provided with two pressure relief valves conforming to Drawing 803-1895224.

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3.3.6 Plastic sheeting. The plastic sheeting used in the construction of the escape appliance hood shall conform to MIL-P-18080 except that the thickness of the sheeting shall be 0.020 inch plus or minus 10 percent.

3.3.7 Color. The color of the escape appliance body, hood, neck seal, and fabric tape shall acceptably match color chip number 12246 of FED-STD-595. The color of the pouch, belt, back pad, and toggle pouch shall acceptably match the dry standard color chip number 26081 of FED-STD-595.

3.3.8 Thread. Thread used for sewing the pouch, webbing, toggle pouch, back pad, accessory pocket, and hood shall conform to type II, size E of V-T-295 and shall be colored to match the items sewn.

3.3.9 Belt hardware. Belt hardware shall conform to Drawing 803-1895220. All belt hardware shall have a lusterless black chemical finish.

3.3.10 Whistle. Each escape appliance shall be provided with a whistle and lanyard. The whistle shall be made of a high quality black plastic with a cork ball. The whistle shall emit a shrill sound clearly audible at a minimum distance of 700 feet. The lanyard shall be made of 1/16 inch diameter commercial braided black nylon cord. The braid shall be 36 inches long and suitably closed to form an approximate 18 inch loop. The lanyard shall be attached to the whistle and to the appliance pocket grommet as indicated on Drawing 803-1895218.

3.3.11 Nose clips. Each escape appliance shall be provided with a nose clip and lanyard conforming to Drawing 803-1895218. The lanyard shall be made of 1/16 inch diameter commercial braided black nylon cord. The braid shall be 20 inches long and closed to form an approximate 10 inch loop. The lanyard shall be attached to the nose clip and to the appliance pocket grommet as indicated on Drawing 803-1895218.

3.3.12 Cement. The adhesive used in the fabrication of the appliance shall be polychloroprene conforming to class 1, 2, or 3 of MIL-A-5540, except that qualification tests are not applicable; the color shall acceptably match the escape appliance coated cloth; the cement used to secure the sealing tape over the slide fastener shall be natural rubber, conforming to class 3, form B of MMM-A-139.

3.3.13 Snorkel valve. Each escape appliance shall be provided with a snorkel valve and a hood relief valve conforming to Drawing 803-1895223.

3.3.14 Inlet check valve. Each escape appliance shall be provided with an inlet valve conforming to Drawing 803-1895222. The air pressure required to open the valve inlet shall be 10 $\pm$ 2 lb/in^2.

3.3.15 Inlet check valve adapter. Each escape appliance shall be provided with an inlet check valve adapter conforming to Drawing 803-1895222.

3.3.16 Slide fastener. The slide fastener shall be nylon on nylon tape with an enameled brass slider and shall conform to type I, style 2, size LM of V-F-106, except that the tape width shall be 9/16 inch. Length of the slide fastener shall be 21 inches. A 1-1/2 inch diameter pull ring shall be attached

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to the slider. The pull ring shall have a round cross section. The fastener shall pass the salt spray test specified in V-F-106. Color of the tape and enamel shall acceptably match the color of the escape appliance (see 3.3.7). All metal components of the slide fastener shall be brass of the composition specified in V-F-106.

3.3.17 Snap fastener. The accessory pocket and cover for the adapter shall each be provided with one snap fastener. The fasteners shall conform to MIL-F-10884 and Drawing 803-1895222.

3.3.18 Press stud. Press stud fasteners used on the escape appliance pouch and toggle line pouch shall conform to Drawing 803-1895220.

3.3.19 Grommets. The grommets used on the accessory pocket and neck seal, as shown on Drawing 803-1895222, shall be plain grommets with plain washers, size "0", brass, black chemical finish.

3.3.20 Reflective tape. The material for the reflective patches shall be SOLAS grade number 3150, 2 inches wide, Minnesota Mining and Manufacturing Company, or equivalent. The primer shall be 2141 Minnesota Mining and Manufacturing, or equivalent.

3.4 General requirements.

3.4.1 Design, manufacture, and assembly. The design, manufacturing, and assembly details of the escape appliance shall conform to Drawings 803-1895218, 803-1895219, 803-1895220, 803-1895221, 803-1895222, 803-1895223 and 803-1895224.

3.4.1.1 Hood. The contractor shall not be required to follow the exact sequence of hood construction as outlined on Drawing 803-1895223.

3.4.2 Cementing. In the construction of the seams, sealing area, and attachment of all component parts of the escape appliance, either cementing or pressure-curing by heat shall be used. In all cementing operations, the surfaces to which the adhesive is to be applied shall be thoroughly washed, if necessary, with a solvent in such a manner that the dusting materials (zinc stearate or talc) and any other surface contaminant are completely removed. The surface shall be clean prior to cementing. Care shall be exercised to ensure that neither the coating nor the base undercloth is damaged, nor that the adhesion between both is impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive and shall leave no residue. Cemented areas shall be rolled and "stitched out", and all trapped air and wrinkles that would impair the function of the escape appliance shall be removed. Adhesives shall be controlled to ensure that old adhesive or adhesive which has partially set up is not used. A fresh batch of adhesive shall be mixed at least every 8 hours during the course of manufacture. Containers for adhesive shall be free from congealed adhesive before being refilled.

3.4.2.1 Cemented area. The adhesive on all outside cemented areas in the way of taped seams and patches shall be applied to the area and shall extend not less than 1/8 inch and not more than 3/8 inch beyond the edges of the tape or patch or parallel thereto.

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3.4.2.2 Cemented seams and sealing areas. The construction of seams and sealing areas and the attachment of all component parts to the escape appliance shall be undertaken utilizing the technique and precautions specified in 3.4.2 so that prior to the submission of the completed escape appliance for final approval all such seams and sealed areas shall develop the optimum properties of the adhesive specified in MIL-A-5540 and MMM-A-139.

3.4.3 Marking. Identification marking shall be by the silk screen method in black letters and numerals 1/4 inch high. The ink shall be black waterproof and cause no deterioration of fabric. A marking patch shall be located on the outside back of the escape appliance as shown on Drawing 803-1895222 and the outside surface of the pouch flap shall be marked as shown on Drawing 803-1895219.

3.4.3.1 Serial numbers. Serial numbers shall be furnished to the successful bidder covering the quantity specified in the contract. Serial numbers shall be located as shown on Drawing 803-1895222. Serial numbers shall not be required to be marked at any other location including the container used for packaging.

3.5 Workmanship. Workmanship shall be such as to ensure compliance with the requirements of this specification. Because of the emergency and life support use of this equipment, it is essential that the escape appliance functions properly under conditions to which it may be subjected.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 First article inspection report. The contractor shall prepare a first article inspection report in accordance with the data ordering document (see 6.2.2).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). The inspection performed on the completed escape appliance and component material shall consist of all the tests specified in this specification. Samples of all materials used in the fabrication of the escape appliance shall be examined and tested to ascertain their conformance to the applicable specifications or to the requirements for the material as specified herein.

4.3.1 Samples for qualification inspection. The following samples shall be submitted for qualification inspection:

(a) Two complete escape appliances.
(b) Two cured pieces each of escape appliance and pouch coating compounds, 6 inches long and 6 inches wide and 0.07 to 0.08 inch thick.
(c) Two yards each of coated and uncoated cloth used in escape appliance and pouch, together with the name and code number of the contractor.
(d) One piece of coated cloth tape 36 inches wide and 72 inches long.
(e) Four coating adhesion samples 12 inches wide and 12 inches long (two samples of the escape appliance coated cloth prepared as specified in MIL-C-19002 and two samples of the coated pouch cloth prepared as specified in MIL-C-17415).
(f) Four cement adhesion samples 7 inches wide and 8 inches long of the coated appliance cloth prepared with the cement used in the appliance (two of the samples prepared coated face to coated face and two prepared uncoated face to coated face).
(g) Five yards of nylon webbing of each width and type used in the appliance.
(h) One 20 by 50 inch sheet of vinyl of the thickness used in the appliance.
(i) One cone (500 yards) of nylon sewing thread of each color used in the appliance.
(j) Four sets of all hardware, including the oral system but excluding the snorkel valve, used in the appliance, together with the name and code number of the contractor.
(k) Two pressure relief valves.

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4.3.1.1 Qualification inspection samples shall be plainly identified by securely attached durable tags marked with the following information:

(a) "Samples for qualification inspection".
(b) Name of product (together with complete identifying data, such as manufacturer's designation or number).
(c) Name of manufacturer.
(d) Submitted by (name), (date) for qualification inspection in accordance with the requirements of MIL-E-23155 under authorization (reference authorizing letter).

4.4 First article inspection. Before starting regular production, the contractor shall submit samples as specified in 4.3.1, except that only one complete escape appliance shall be required. Samples shall be subjected to the examinations and tests of 4.6.1 and 4.7.

4.4.1 First article test samples shall be forwarded to Commander, David Taylor Naval Ship Research and Development Center, Annapolis, MD 21402, Attention: Code 2843, plainly identified by securely attached durable tags marked with the following information:

(a) "Samples for first article test."
(b) Name of product, together with complete identifying data, such as manufacturer's designation number.
(c) Name of manufacturer.
(d) Contract or order number.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed by the contractor under the surveillance of the Government. The inspection shall consist of the examination specified in 4.6.1 and the tests specified in 4.5.1. The results of the pressure, leakage, and buoyancy tests shall be identifiable by listing the assigned serial numbers. To ensure that the final product is sound and that it conforms to the requirements of this specification, the contractor shall provide for the adequate control of the quality of the coated fabric, the adhesive, and the attachments during the process of manufacture.

4.5.1 Sampling for quality conformance inspection.

4.5.1.1 Lot. A lot shall consist of all life preservers submitted for inspection at one time.

4.5.1.2 Sampling for examination. A random sample of life preservers shall be selected from each lot in accordance with inspection level I of MIL-STD-105 for the examination specified in 4.6.1. The acceptance number shall be zero for critical defects. The acceptable quality level (AQL) shall be 4.0 for major defects and 10.0 for minor defects per 100 units.

4.5.1.3 Sampling for tests. A random sample of life preservers shall be selected from each lot in accordance with the following inspection levels as specified in MIL-STD-105 for the tests specified:

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Tests	Inspection level
Operation of relief valve (see 4.8.4)	I
Pressure (see 4.8.2)	I
Operation of oral valve (see 4.8.1)	I
Operation of air inflation system (see 4.8.5)	I
Leakage (see 4.8.3)	I
Buoyancy (see 4.8.6)	S-3

4.5.1.3.1 The acceptance number shall be zero for all tests. Rejected escape appliances shall not be resubmitted.

4.5.1.3.2 The following tests shall be conducted as shown below on each escape appliance by the contractor under the surveillance of the inspector in the order listed. Escape appliances not meeting the requirements of these tests shall be corrected prior to submitting the escape appliances for acceptance, but under no circumstances shall an air retaining seam be reworked more than one time.

Tests	Quantity	Paragraph
Pressure	Each	4.8.2
Leakage	Each	4.8.3
Operation of relief valves	Each	4.8.4
Operation of oral valve	Each	4.8.1
Operation of air inflation system	Each	4.8.5
Oral inflation tube (porosity)	Each	4.8.7.1
Oral inflation tube (tensile)	6 samples per 1000	4.8.7.2
Oral inflation valve adhesion	6 samples per 1000	4.8.10

4.5.1.3.3 In addition, the contractor shall prepare a minimum of 10 adhesion test samples from each batch of adhesive and shall subject the samples to the test specified in 4.8.8. The dimensions of the test specimen shall be in accordance with the requirements for "strip" specimens for the test in accordance with method number 8011 as specified in FED-STD-601.

4.6 Comparison tests. NAVSEA may conduct, at a Government laboratory, comparison tests on random samples of escape appliances produced under any contract. Comparison tests may include any or all of the tests of this specification. Acceptance and shipment of escape appliances shall not be delayed pending completion of comparison tests. However, if any comparison test sample is rejected for noncompliance with the requirements of this specification, acceptance and shipment of any further escape appliance shall be withheld until the contractor can demonstrate that he has corrected the conditions which led to the rejection of the comparison test sample. Samples selected for comparison tests shall be considered a part of the quantity to be delivered under the contract and shall be returned with all applicable test reports, recommendations, and comments pertinent for monitoring production.

4.6.1 Examination. Sample escape appliances selected in accordance with 4.5.1.2 shall be thoroughly examined to determine conformance to this specification and the applicable drawings. Examination shall be conducted as specified in table III.

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TABLE III. Classification of defects.

Categories	Defects
Critical:	
1	Buoyancy chamber material defective; evidence of splits, patch holes, tears, fraying.
2	Buoyancy chamber seams defective, incomplete, or not as specified; evidence of bubbles, improperly applied tape, incomplete cementing, or seams in buoyancy chamber not the specified type; width of seams less than the required minimum.
3	Oral inflation tube defective, not secure, securing patches not as required, or diameter of base not as specified.
4	Oral inflation valve defective or not secure in tube; tube evidences cracks, splits, bubbles, bulges, pin holes, or other deleterious effects.
5	Protective tape not removed from inside of relief valve prior to assembly.
6	Pressure relief valves missing, defective, not as required, not properly installed and located.
7	Hood plastic sheet material defective; evidence of blisters, holes, splits, or cloudiness.
8	Inlet check valve defective, not as required, not properly installed and located.
9	Hood incomplete or not as specified, improperly attached to appliance, incomplete cementing.
Major:	
101	Pouch defective, not securely attached as required, or stitching incomplete; press stud fasteners defective or not secured as required for pouch and harness.
102	Belt evidences tears, splits, fraying, or weak spots.
103	Belt stitching defective, incomplete, or not as specified.
104	Belt hardware missing, defective, or not secure; evidence of cracks, deep pits, thin spots, or sharp surfaces; buckle cannot be engaged or length of belt not adjustable.
105	Belt length not as required.
106	Toggle line defective or toggle not secured.
107	Metal components not plated or finished as specified; plating or finish defective or incomplete.
108	Date of manufacture marking missing, illegible, or incorrect.
109	Light attachment loop defective or not attached as specified.
110	Evidence of reworking air retaining seams more than the allowable number of times.
111	Adhesive smears on plastic face piece.
112	Reflective tape shows evidence of tears or splits; tape not cemented as required.

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TABLE III. Classification of defects. - Continued

Categories	Defects
Minor:	
201	Length of toggle line not as required.
202	Color of cloth, webbing, thread, and adhesive not as specified.
203	Marking, manufacturer's name or trademark, serial number, or identification missing, incorrect, illegible, not permanent, or not located as specified.
204	Escape appliance not folded in pouch prior to packing.
205	Packaging and packing not as specified.
206	Adhesive on fabric surface around accessory patches and tapes below the minimum limit of 1/8 inch and in excess of 3/8 inch.

4.6.2 Resubmitted lots. Resubmitted lots (allowed in connection with major and minor defects found in the examination) may be resubmitted for examination as specified in 4.5.1.2. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be selected with the acceptance number remaining zero. The resubmitted lot shall be examined by the contractor under the surveillance of the Government inspector using tightened examination. Before a lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the lot shall be furnished by the contractor to the Government inspector.

4.7 Test conditions.

4.7.1 Atmospheric conditions. Unless otherwise specified in the contract or order, all tests required by this specification shall be conducted at prevailing atmospheric pressure and a temperature of 77 $\pm$ 18 degrees Fahrenheit (°F) and a relative humidity of 80 percent or less. If the final values of the ambient temperature or barometric pressure at the end of the 24 hour air pressure tests are different from the initial values, the following corrections shall be made to the final pressure readings:

(a) Temperature corrections: For each degree rise in temperature, 0.031 lb/in^2 shall be subtracted from the final pressure reading. For each degree drop in temperature, 0.031 lb/in^2 shall be added to the final pressure reading.
(b) Barometric pressure correction: For each 0.1 inch of mercury (Hg) rise in barometric pressure, 0.049 lb/in^2 shall be added to the final temperature corrected pressure reading. For each 0.1 inch of Hg drop in barometric pressure, 0.049 lb/in^2 shall be subtracted from the final temperature corrected pressure reading.
(c) Pressure shall be measured by means of a Hg manometer and then converted to lb/in^2 by multiplying inches of Hg by 0.49.

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4.8 Test conditions.

4.8.1 Operation of oral valve. The escape appliance shall be inflated through the oral valve by means of a blow gun having a discharge nozzle not exceeding 1/8 inch diameter when the gun is connected to a 90 ± 5 lb/in^2 pressure source. The preserver shall be inflated until the relief valves open. The oral valve shall work without difficulty and after the test shall not leak under water with the knurled nut in the opened position and shall shut off securely.

4.8.2 Pressure. With both relief valves capped or sealed off, the escape appliance shall be inflated with air through the oral inflation valve or through the air inflation system to a pressure of 5 lb/in^2 and shall be allowed to stand for a minimum of 5 minutes. The pressure drop shall not be greater than 0.4 lb/in^2. The escape appliance shall be examined while held under pressure. There shall be no evidence of failure of the material in any respect. All seams and sealed areas shall remain perfectly intact and shall show no signs of separation.

4.8.3 Leakage. With both relief valves capped or sealed off, the escape appliance shall be inflated with air through the oral inflation valve or through the air inflation system to a pressure of 1 lb/in^2 and allowed to stand for a minimum of 15 minutes. The pressure shall then be checked and readjusted if necessary to the original pressure of 1 lb/in^2. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 24 hours after the readjustment period the pressure shall be measured and corrected for any change in temperature or barometric pressure (see 4.7). The corrected pressure shall be not less than 0.75 lb/in^2. The oral inflation system shall be examined when the compartment is being deflated to determine that the valve works without difficulty and shuts off securely.

4.8.4 Operation of relief valves. The apparatus shall be inflated with air to test the operation of each relief valve. Each valve shall relieve at 1.50 ± 0.25 lb/in^2 and shall close at not less than 1 lb/in^2.

4.8.5 Operation of air inflation system. The escape appliance shall be inflated through the high pressure air system by means of a quick-disconnect coupling, Schrader X19011B-197, or equal. The pressure within the escape appliance during a 1-minute inflation shall not exceed 2.75 lb/in^2 when the coupling is connected to a 1/4 inch inside diameter air line with a 90 ± 5 lb/in^2 pressure source. The high pressure inflation system shall work without difficulty and after the test shall not leak. The quick-disconnect coupling shall engage and disengage from the high pressure inflation system without difficulty.

4.8.6 Buoyancy. The escape appliance shall be inflated with air until the relief valves open. The valves shall then be capped or sealed off. The escape appliance shall then be attached to a steel weight, the latter weighing 29 pounds when submerged in fresh clean water. With hood closure unfastened, the escape appliance shall support this 29 pound weight without sinking.

4.8.7 Oral inflation tube. Before the oral inflation valve is clamped in the oral inflation tube, the tests specified in 4.8.7.1 and 4.8.7.2 shall be conducted.

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4.8.7.1 Porosity. The oral inflation tube shall be subjected to an air pressure of 25 lb/in^2 for a minimum of 10 seconds while submerged under water. There shall be no evidence of air leakage due to porosity. There shall be no evidence of weakening of the tube as revealed by excessive deformation in any portion of the tube.

4.8.7.2 Tensile. The oral inflation tube base shall be folded longitudinally so that the rounded edges coincide and then placed in the lower jaws of a test machine. The tube portion shall be placed in the upper jaws. The rate of load application shall be 12 ± 1/2 inches per minute. The tensile strength shall be not less than 75 pounds.

4.8.8 Adhesion. The test for adhesion shall be conducted in accordance with method 8011 specified in FED-STD-601. The adhesive strength of the cemented samples shall be not less than 7.5 pounds per inch of width for samples joining the coated sides of the finished fabric and not less than 5.0 pounds per inch of width for seams joining coated side to uncoated side of the finished fabric except as indicated in 3.4.2. The adhesion of the tape seal slide fastener cover, coated side of tape to uncoated side of appliance, shall be a maximum of 5.0 pounds per inch of width.

4.8.9 Salt spray. A 100 hour salt spray shall be conducted on the oral valve, relief valve, snorkel, slide fastener, and complete air inflation system in accordance with ASTM B 117. There shall be no evidence of corrosion which will adversely affect the operation of the equipment.

4.8.10 Oral inflation valve adhesion. This test shall be conducted after the oral inflation valve is clamped within the oral inflation tube. The oral inflation tube base shall be folded longitudinally so that the founded edges coincide and then placed in the lower jaws of a machine. The oral inflation valve shall be placed in the upper jaws. The rate of load application shall be 12 ± 1/2 inches per minute. The load to separate the oral inflation valve from the oral inflation tube shall be not less than 50 pounds.

4.8.11 Hood leakage. With the slide fastener sealed with tape, the snorkle and the hood relief valve shall be sealed off to prevent water leakage. The hood shall be filled with water up to but not including the slide fastener and allowed to stand for 5 minutes. There shall be no evidence of leakage at the end of the test.

4.9 Possible test failures. Possible test failures shall be determined as follows:

(a) Operation of assembled escape appliance.
  (1) Escape appliance cannot be readily inflated through the high pressure air inflation system or through the oral inflation valve.
  (2) Air leaks back through the oral valve or high pressure air system after inflation.

(b) Operation of relief valve:
  (1) Relief valves fail to relieve within the specified limits when inflated as required.

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(c) Pressure test:
(1) Evidence of material failure or parting of seams when escape appliance is subjected to the specified pressure test.
(d) Buoyancy:
(1) Failure of the inflated escape appliance to support, without sinking, the specified weight.

4.11 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Domestic shipment and early use.

5.1.1 Packaging. Packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early use and may conform to the contractor's commercial practice when such meets these requirements.

5.1.2 Packing. Packing shall be accomplished in a manner which will ensure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name, and destination.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of packaging, packing, and marking for shipment shall be specified by the contracting activity (see 6.2.1).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when acquisition is made.

5.2.1.1 Packaging, packing, and marking. Escape appliances shall be packaged level A or C, packed and marked level A, B, or C as specified (see 6.2.1) in accordance with MIL-L-3454, except that prior to packing, escape appliances shall be folded in pouches (see 5.2.1.1.1).

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5.2.1.1.1 Folding. The following procedure shall be followed for packing the appliance into the pouch:

(a) Deflate appliance and lay flat. Set snorkel valve in open position. Cover snorkel with scrap coated fabric. Push snorkel inward to hood. Fold sides of appliance in 2 to 3 inches. Do not bend or crease oral tube.
(b) Flatten out hood. Press plastic face inward (dish) without creasing plastic. Place lower part of appliance into pouch. Roll down from top of appliance into pouch.

6. NOTES

6.1 Intended use. Escape appliances made in accordance with this specification are intended for use on board submarines.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article inspection (see 3.2).
(c) Packaging, packing, or marking requirements other than those required by 5.1 (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2	First article inspection report	DI-T-4902	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12., Vol. I, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-23155 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of the samples specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Escape appliance
Pouch
Valve, pressure, relief

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4220-N317)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-23155C(SH)	Escape Appliance, Hooded, Air Inflatable, With Pouch, Submarine Personnel

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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